

# Applied Thermodynamic II

## ME2402

# *Psychometric & Air-conditioning*



# ITUM

INSTITUTE OF TECHNOLOGY  
UNIVERSITY OF MORATUWA

DIVISION OF MECHANICAL  
ENGINEERING TECHNOLOGY

*Conducted by Chathura Jayasekara*

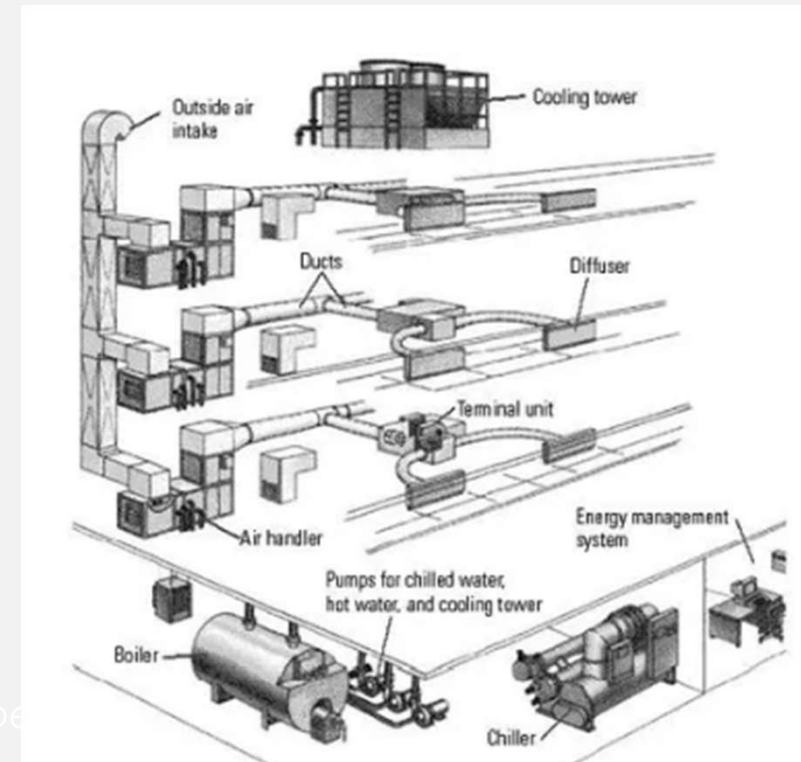
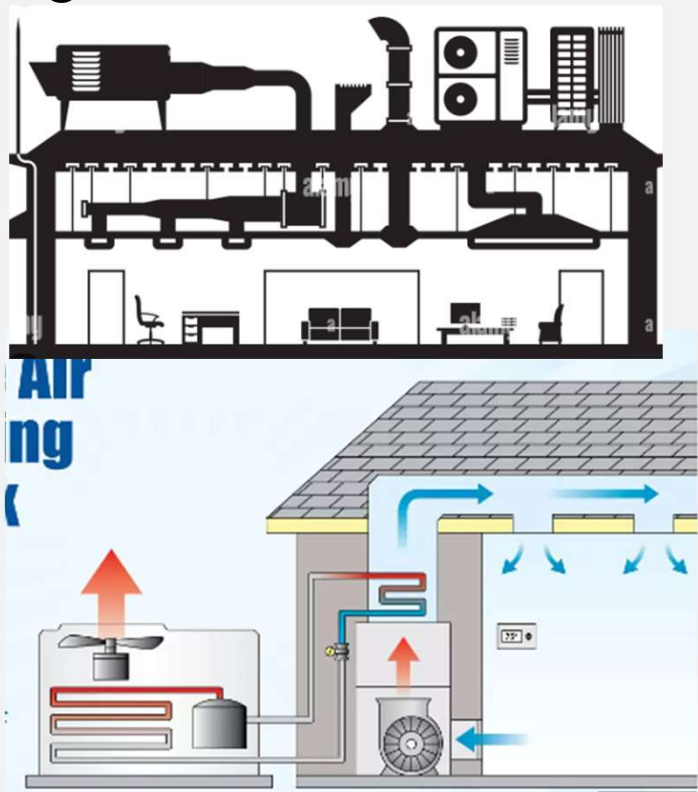
## 3.0 Air conditioning System

- Air conditioning is a application of **Refrigeration Cycle** in a large scale
- Further there are certain **requirements** to fulfill
  - i. Should maintain the temperature
  - ii. Should maintain the Relative Humidity
  - iii. Maintain the air purity level
  - iv. Maintain the required level of air circulation

<https://www.youtube.com/watch?v=jrWHi77oAWI>

### 3.0 Air conditioning System

So, the air-conditioning applications vary from span of few cubic meters to higher amount



<https://www.youtube.com/watch?v=...>

## 3.0 Air conditioning System

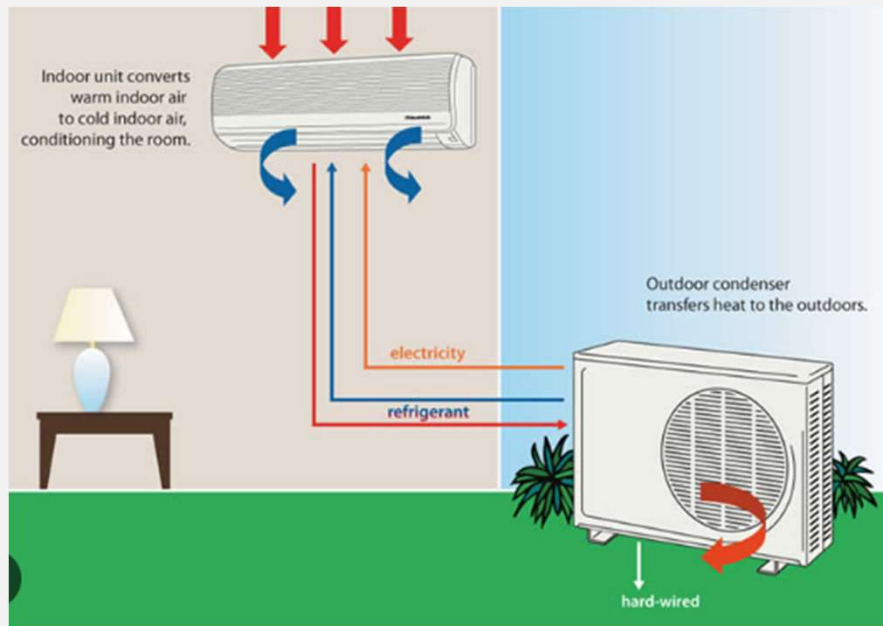
Air-conditioning systems can be categorized as follows

- i. Single or direct air conditioning
- ii. Central Air conditioning : Chill water cooling systems

<https://www.youtube.com/watch?v=jrWHi77oAWI>

## 3.0 Air conditioning System

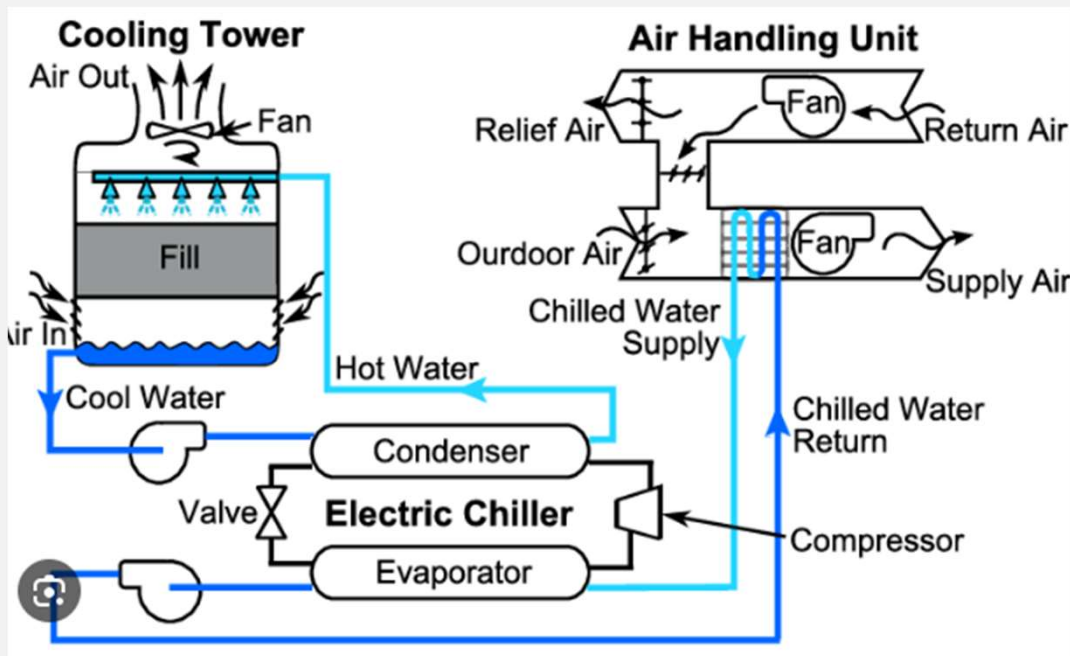
### i. Single or direct air conditioning



<https://www.youtube.com/watch?v=jrWHi77oAWI>

## 3.0 Air conditioning System

### ii Central Air conditioning chill water system



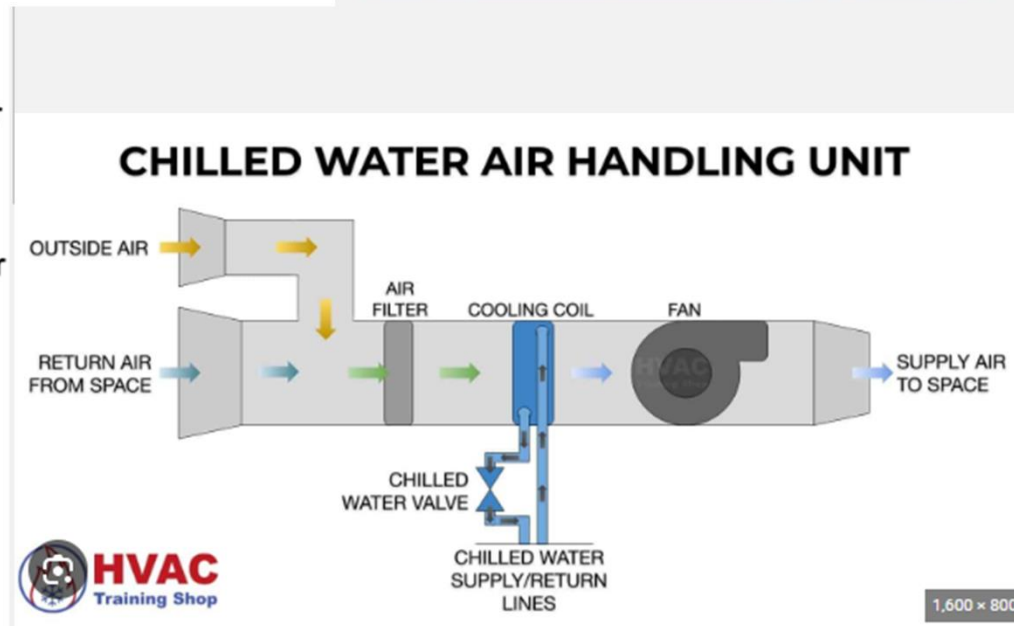
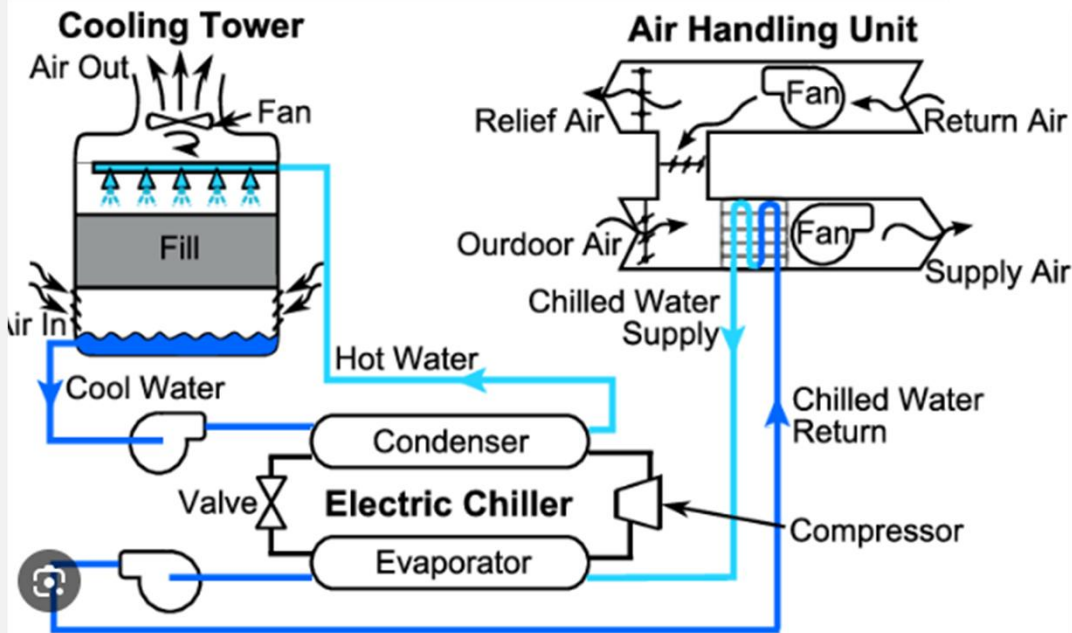
There are three circulation systems in this

- i. Refrigeration circuit (cycle)
- ii. Evaporator water circuit
- iii. Condenser water circuit

<https://www.youtube.com/watch?v=jrWHi77oAWI>

## 3.0 Air conditioning System

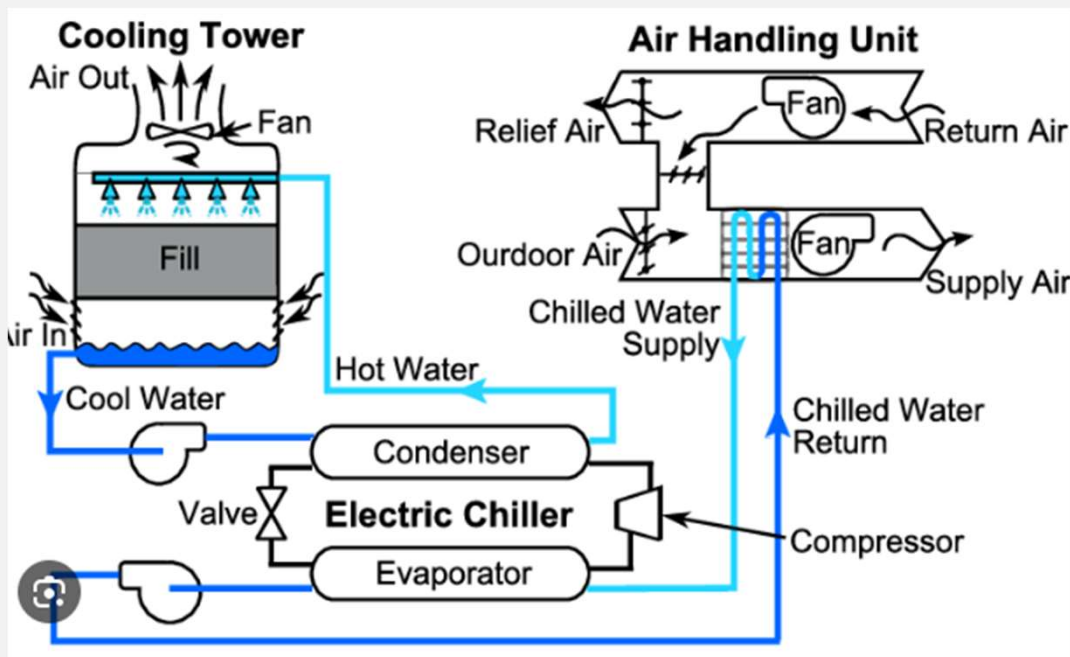
### ii Central Air conditioning chill water system



<https://www.youtube.com/watch?v=jrWHi77oAWI>

## 3.0 Air conditioning System

### ii Central Air conditioning chill water system



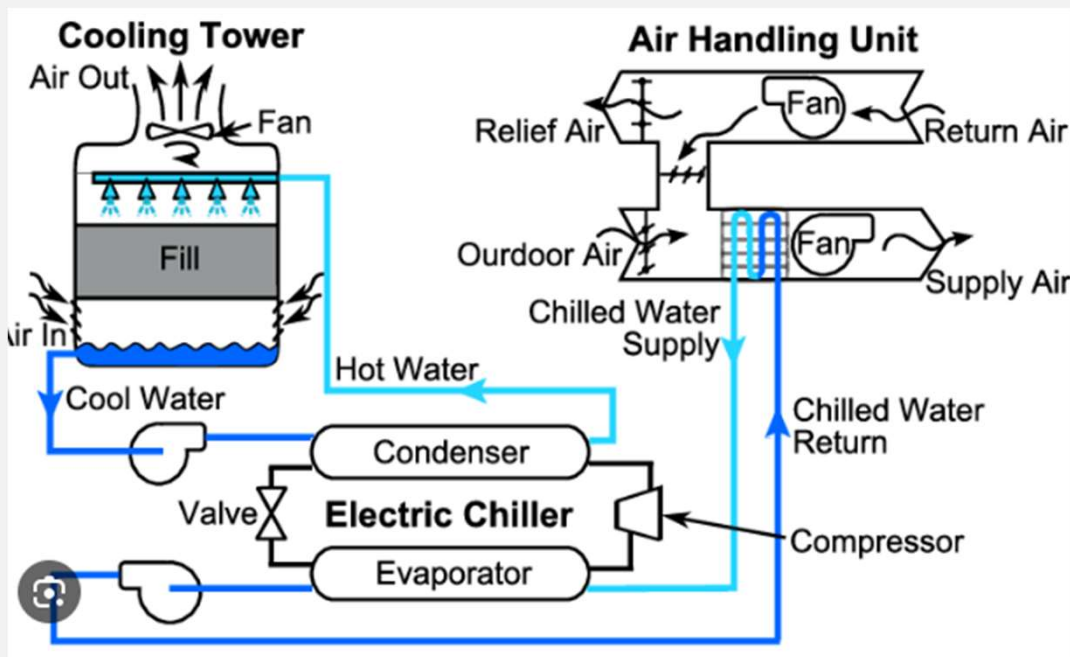
### Refrigeration cycle is in the chiller plant

- Refrigerant cycle provide the cooling effect
- Absorb heat from evaporator chill water
- Then deliver condenser cooling water at condenser

<https://www.youtube.com/watch?v=jrWHi77oAWI>

## 3.0 Air conditioning System

### ii Central Air conditioning chill water system



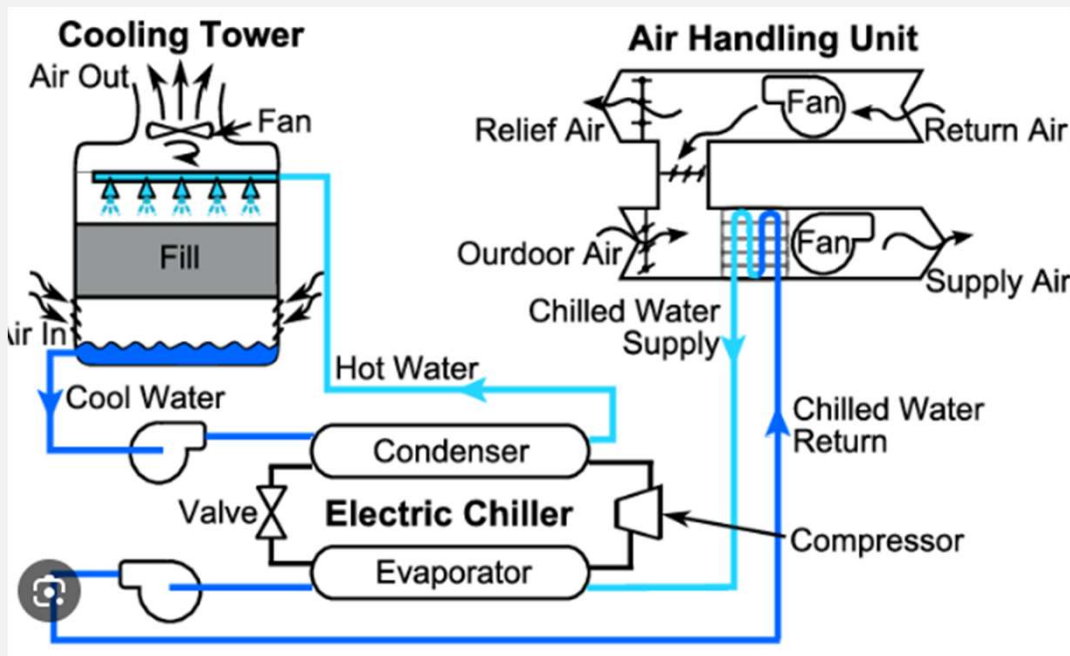
### Evaporator water circuit

- Use water circuit over the evaporator heat exchanger.
- Heat absorbed from the space is delivered to refrigerant at the evaporator by chilled water

<https://www.youtube.com/watch?v=jrWHi77oAWI>

## 3.0 Air conditioning System

### ii Central Air conditioning chill water system



### Condenser water circuit

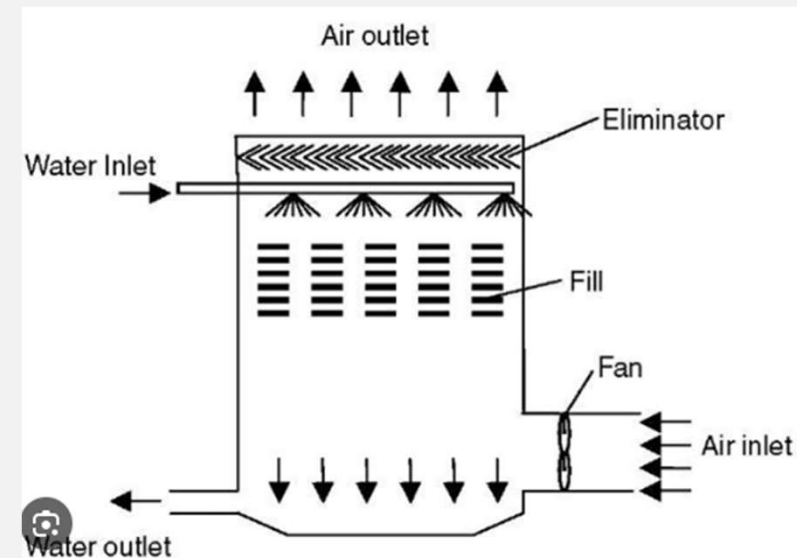
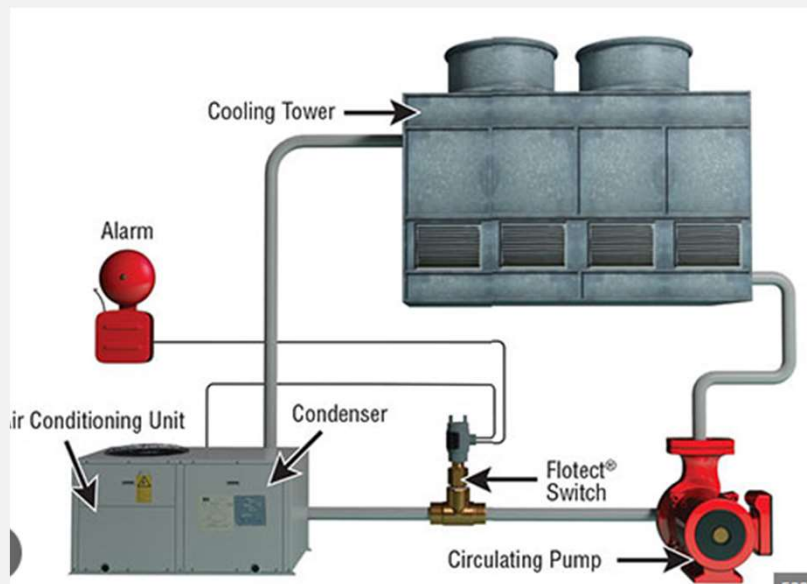
- Use water circuit over the condenser heat exchanger.
- Heat carried in the refrigerant is delivered to the condenser water
- Condenser water remove this heat to the atmosphere at a cooling tower or radiator

<https://www.youtube.com/watch?v=jrWHi77oAWI>

## 3.0 Air conditioning System

### ii Central Air conditioning chill water system

#### Condenser water circuit -Cooling tower or radiator

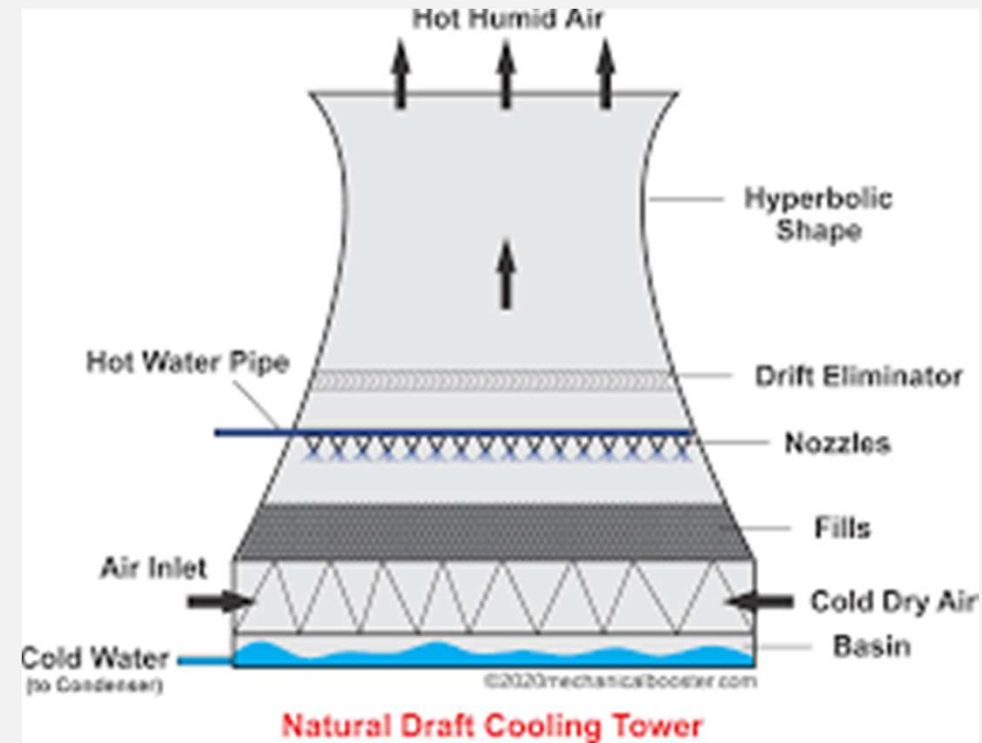


<https://www.youtube.com/watch?v=jrwn1770AwI>

## 3.0 Air conditioning System

### Cooling Towers

- Natural draft cooling towers

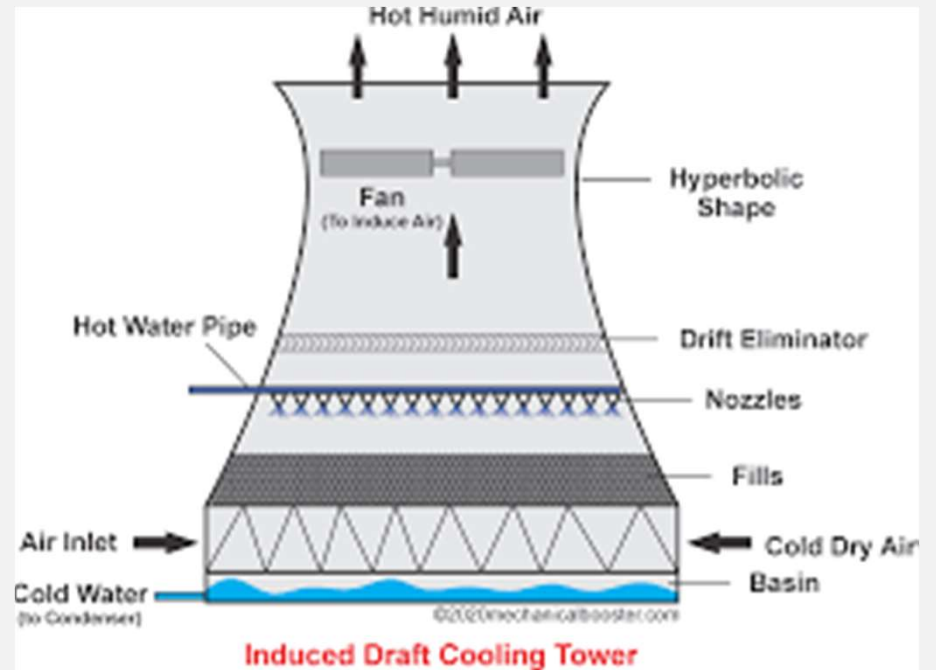


<https://www.youtube.com/watch?v=jrWHi77oAWI>

## 3.0 Air conditioning System

### Cooling Towers

- Induced draft cooling towers



<https://www.youtube.com/watch?v=jrWHI77oAWI>

## 3.0 Air conditioning System

- ii Central Air conditioning  
chill water system

**Condenser water circuit -Cooling tower or radiator**



<https://www.youtube.com/watch?v=VRCOOLER>

## 3.0 Air conditioning System

### ii Central Air conditioning chill water system

- Chilled water is directly sent to the cooling space **OR**
- Air will be chilled at AHU and then chilled air will be circulated to the rooms or the space need to cool
- There need terminal units to allow proper delivery or cool air flow



<https://www.youtube.com/watch?v=jrwn770XvM>

## Air conditioning load

- The air conditioning load is handled by the refrigeration cycle
- Based on the load the capacity of the refrigeration system should be selected
- Air-conditioning load has two part
  - Sensible heat load : energy added at unit time to **increase dry bulb temperature**
  - Latent Heat load : energy added at unit time due to enthalpy of **moisture and evaporation energy**

**So it is necessary to analyze the theory behind air-conditioning**

<https://www.youtube.com/watch?v=jrWHi77oAWI>

# Understanding theory for air-conditioning

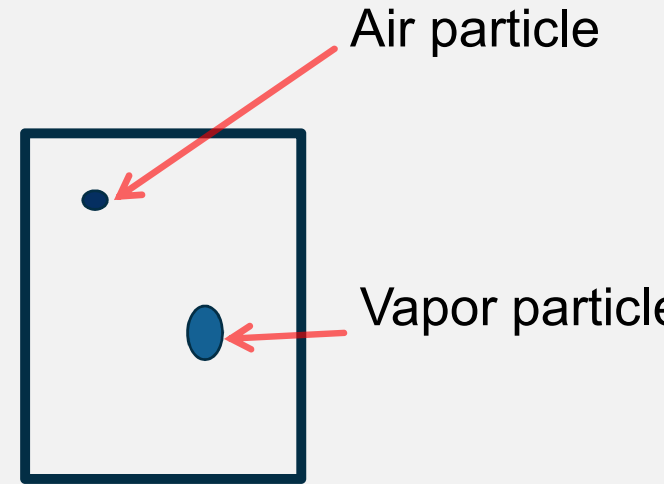
## Composition of atmospheric air

- Typical unsaturated space is having air(dry) and vapor
- This vapor is called super heated vapor
- Both of these components contribute to the total pressure

$$P = P_a + P_s$$

There is amount of vapor (Vapor pressure) that can have in a certain **unit space**

- It depends on the temperature (T)
- It depends on the pressure (P)

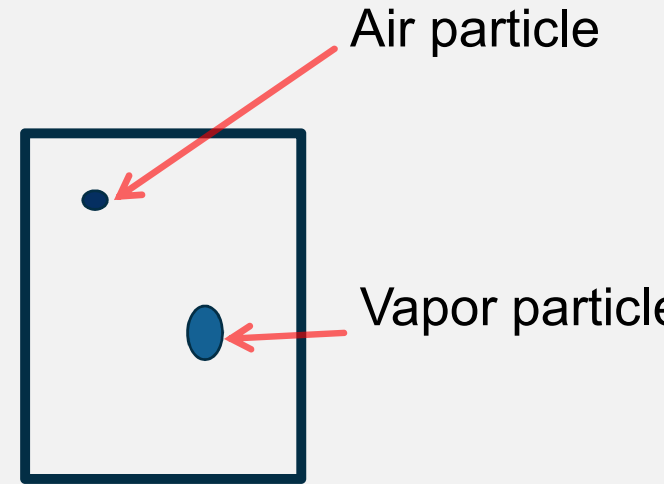


# Understanding theory for air-conditioning

## Composition of atmospheric air

If the limit of vapor content is reached (or pressure reached) then that space cannot hold any more vapor

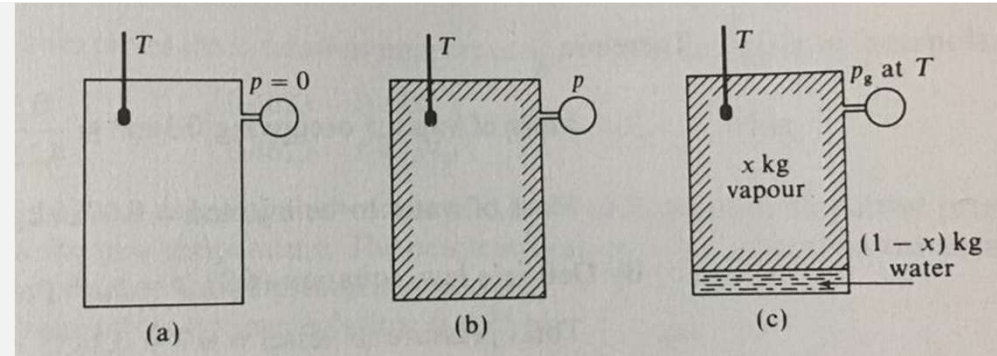
- That means water will not evaporated any more
- Liquid form will remain



## Understanding theory

### Adding water to a space

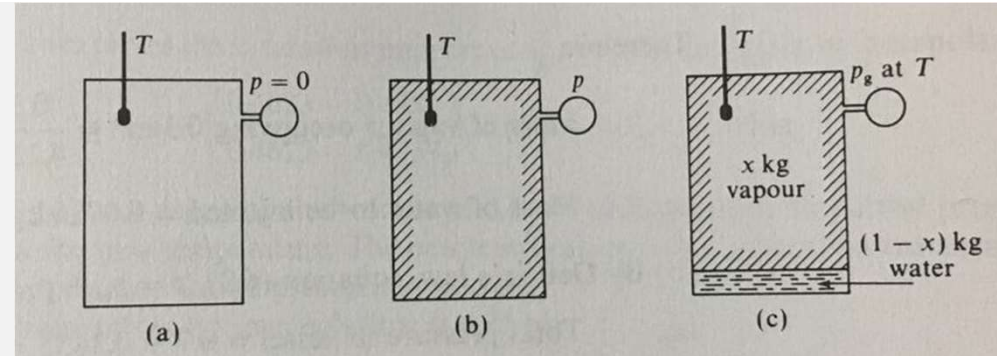
- Consider a unsaturated space as in the figure
- It has air and vapor
- We add more water, it will get evaporated and increase partial pressure vapor ( $p_s$ )
- So, liquid water will not visible
- This will continue **until the maximum amount** of vapor that volume can hold.
- Other words, unit the saturation maximum pressure reach



## Understanding theory

### Adding water to a space

- If further water is added, it will remain as water and will not evaporate.
- Now vapor is in contact with liquid water
- So, if 1kg water is introduced;
  - $x$  kg of dry saturated vapor and  $1-x$  kg of water
  - Or else, can say 1kg of **wet steam with  $x$  dryness fraction**.



# Psychrometric Mixtures

## **Condensation**

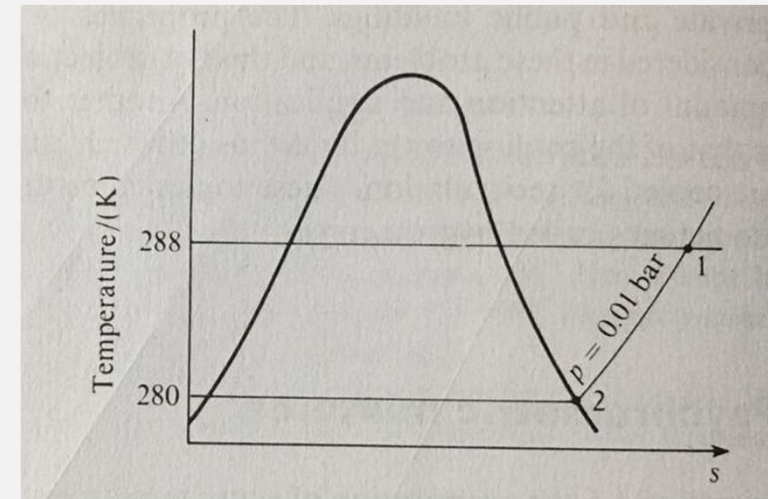
- Consider atmospheric air at 1.013bar and 15 degree C
- The saturation pressure of water vapor at this condition is 0.017bar ~0.01 bar
- The saturation temperature for 0.01bar is 7 degree C
- Now the degree of superheat is 15-7 degree C

## 2.0 Psychrometric Mixtures

### Condensation - application

- Metal cup is having water and in atmosphere
- Put ice cubes and cooled the water continuously
- At a certain temperature, you will see water condensation outside the metal cup
- From previous theory now the vapor in contact with metal cup should be having 7 degree C
- This is the condition of saturation is the **dew point** of mixture given by  $t_D$

*Eg : Mist appear on windscreen on cold days*



## 3.0 Specific Humidity, Relative Humidity and Percentage Saturation

### Specific Humidity $\omega$

- Is the moisture content
- Is ratio of mass of water vapor to mass of dry air, in a volume

Specific Humidity =  $\omega$

$$\omega = \frac{m_s}{m_a}$$

OR

$$\omega = 0.622 \times \frac{p_s}{p - p_s}$$

Here,

$m_s$  is mass of superheated vapor

$m_a$  is mass of dry air

as

$p_s$  is partial pressure of super heated vapor

$p$  is total pressure

## 3.0 Specific Humidity, Relative Humidity and Percentage Saturation

### Relative Humidity $\phi$

- Ratio of the actual mass of water vapor in a given volume to saturated vapor mass
- Is ratio of mass of water vapor to mass of dry air, in a volume

Relative Humidity =  $\phi$

$$\phi = \frac{m_s}{(m_s)_{sat}}$$

OR

$$\phi = \frac{p_s}{p_g}$$

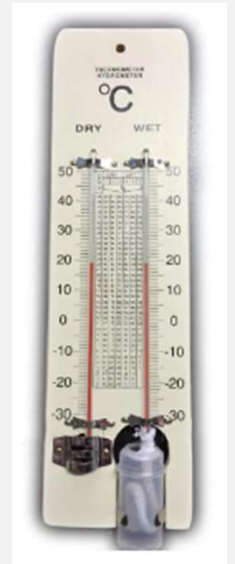
Here,

$m_s$  is **actual mass** of water vapor

$(m_s)_{sat}$  is mass to saturate **same space** at the **same temperature**

$p_s$  is partial pressure of **super heated vapor**

$p_g$  is **saturation pressure** at the **same pressure**



## 3.0 Specific Humidity, Relative Humidity and Percentage Saturation

### Percentage saturation $\varphi$

- Relative humidity can be expressed as percentage saturation
- Percentage saturation is ratio of specific humidity to specific humidity when saturated

Percentage saturation =  $\varphi$

$$\varphi = \frac{\omega}{\omega_g}$$

## 4.0 Psychometric chart

35.

30/11/21

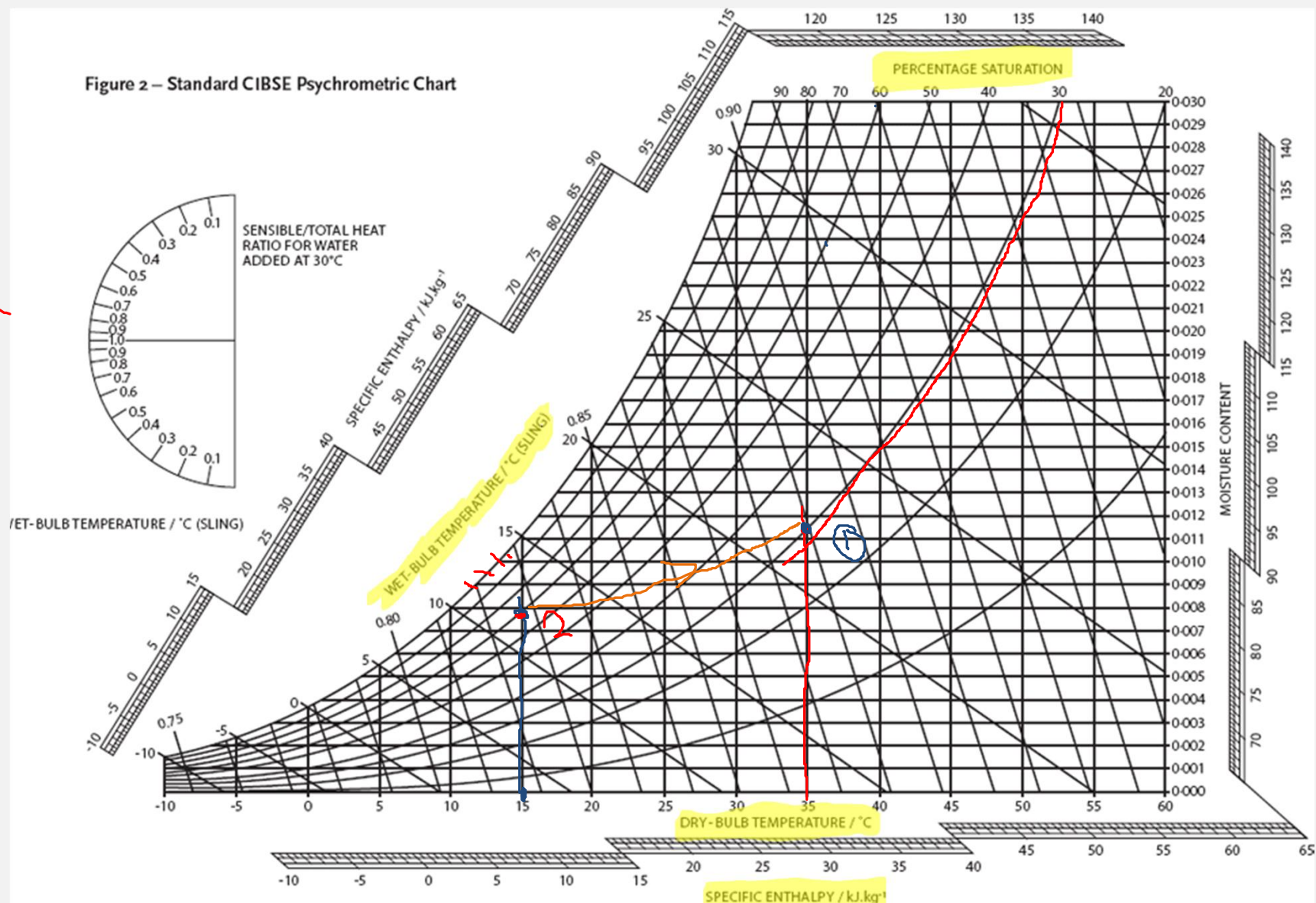
Ne

15°

RA

70%

Figure 2 – Standard CIBSE Psychrometric Chart

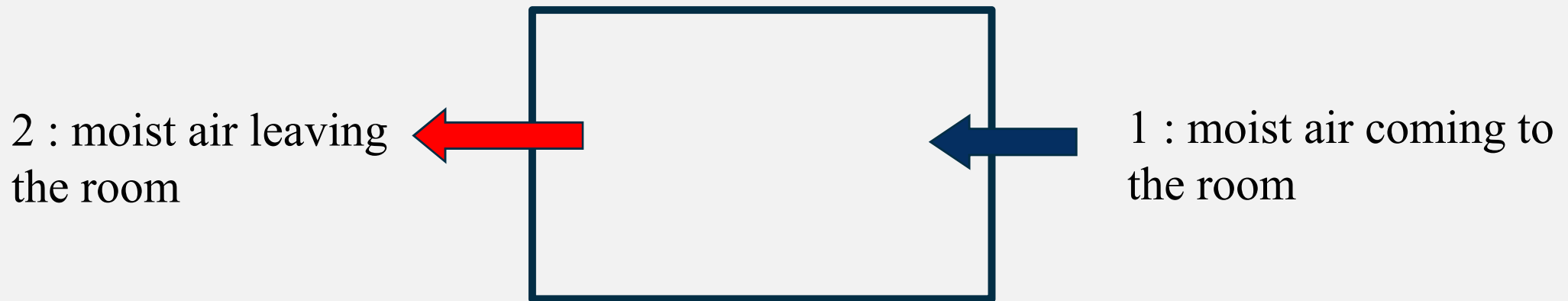


## 5.0 Air-conditioning on psychometric chart

### Air-conditioning

Air-conditioning load has **two parts**

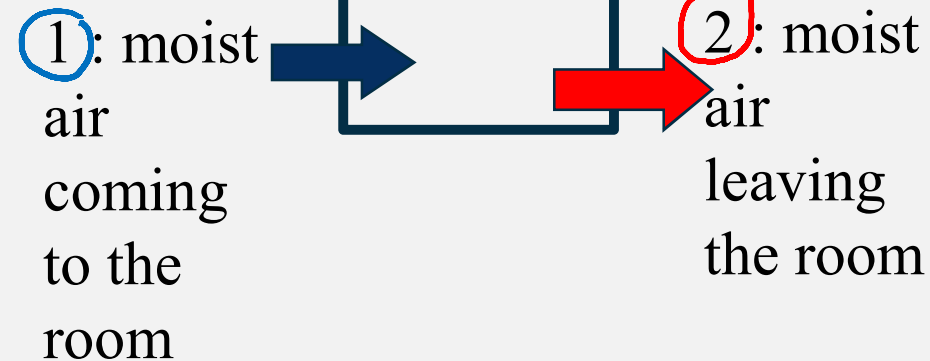
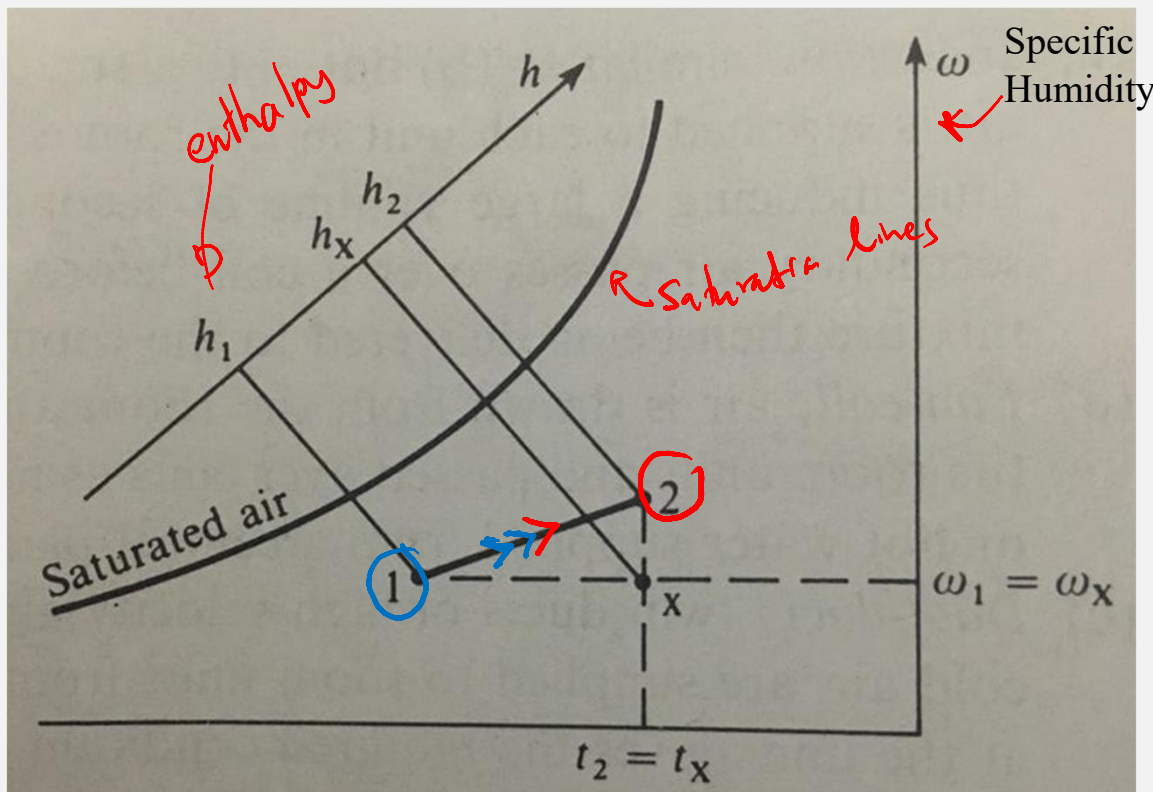
- i. Sensible heat load : energy added to **change dry bulb temp**
- ii. Latent heat load : energy added due to **enthalpy of moisture** + heat required to **evaporated moisture**



## 5.0 Air-conditioning on psychrometric chart

### Air-conditioning process in chart

Above process 1 to 2 can be marked on a psychrometric chart as follows



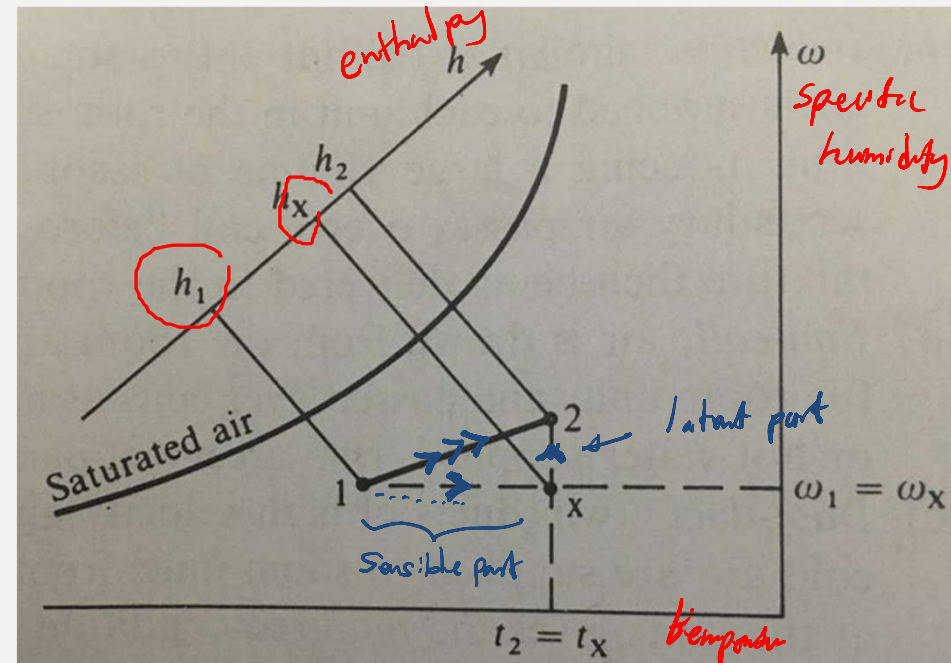
## 5.0 Air-conditioning on psychrometric chart

## Air-conditioning process in chart

Point X is selected such that  $\omega_1 = \omega_x$  and  $t_2 \neq t_x$

$$\text{Sensible heat load} = \dot{m}_a(h_x - h_1) = \dot{m}_a(c_{pa} + \omega c_{ps})(t_x - t_1)$$

mass flow rate of air



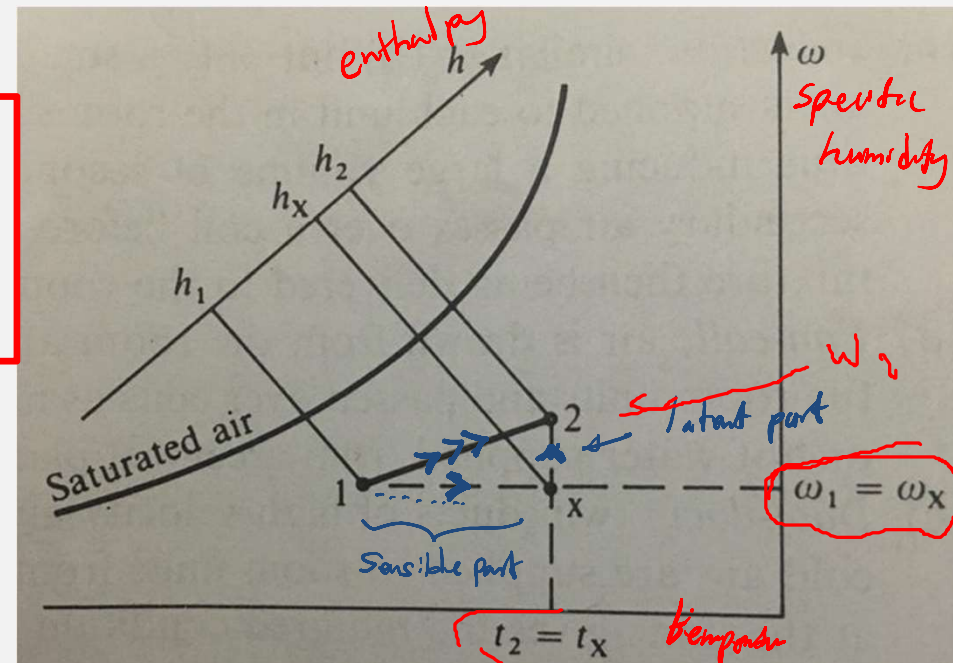
## 5.0 Air-conditioning on psychrometric chart

### Air-conditioning process in chart

Point X is selected such that  $\omega_1 = \omega_x$  and  $t_2 = t_x$

$$\text{Latent heat load} = \dot{m}_a(h_2 - h_x) = \dot{m}_a(\omega_2 - \omega_1)(h_g \text{ at } t_2)$$

$$\begin{aligned} \text{Total heat load} &= \text{Sensible heat} + \text{Latent heat} \\ &= \dot{m}_a(h_2 - h_1) \end{aligned}$$



The air supplied to a room of a building in winter is to be at  $17^\circ\text{C}$  and have a percentage relative humidity of 60%. If the barometric pressure is 1.01326 bar, calculate the specific humidity. What would be the dew point under these conditions?

Solution

$$\boxed{\begin{matrix} 17^\circ\text{C} \\ \text{RH} = 60\% \end{matrix}} \text{ air}$$

$$P_{\text{total}} = P = 1.01326 \text{ bar}$$

Here you can see that it is a temperature changing

type process.

Find the  $P_g$  value at temperature  $17^\circ\text{C}$ ; it is  $P_g = 0.01936$  bar

$$\phi = \text{Relative Humidity} = \frac{P_s}{P_g}$$

$$0.6 = \frac{P_s}{0.01936}$$

$$P_s = 0.011616 \text{ bar}$$

$$\text{the } w = 0.622 \times \frac{P_s}{(P - P_s)}$$

$$(i) w = \frac{0.622 \times 0.011616}{(1.01326 - 0.011616)}$$

$$= 0.007213$$

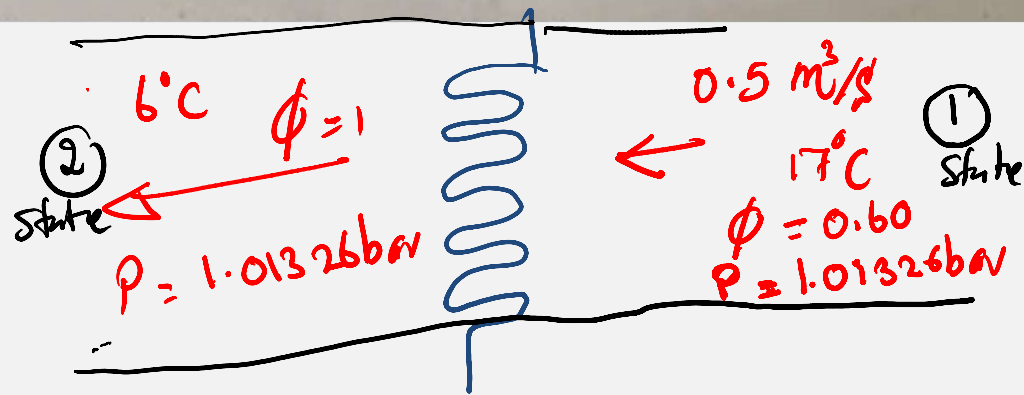
$$(ii) \text{ this means } w = \frac{m_s}{m_a}$$

Per 1 kg of this you have 0.007213 kg of this

find this from table.

now if you cool the system at constant Pressure the system will saturate at a certain temp called (Dew point)

If air at the condition of Example ~~15.1~~ <sup>previous one</sup> is passed at the rate of  $0.5 \text{ m}^3/\text{s}$  over a cooling coil which is at a temperature of  $6^\circ\text{C}$ , calculate the amount of vapour which will be condensed. Assume that the barometric pressure is the same as in Example 15.1, and that the air leaving the coil is saturated.



$$P_a = P - P_s$$

$$\therefore P_a = 1.01326 - 0.011616 \text{ bar}$$

$$P_a = 1.00163 \text{ bar}$$

$$\therefore \dot{m}_a = \frac{10^3 \times 1.00163 \times 0.5}{10^3 \times 0.287 \times 290} = 0.6017 \text{ kg/s}$$

We assume the air flow rate is constant

$$\dot{m}_a = \frac{P_a V}{R_a T}$$

Previously found that  $\omega = 0.007213$

$\omega = \frac{m_s}{m_a}$  is applied for state ①

$$50 \text{ kg}_s = 0.007213 \times \dot{m}_a$$

Here you can see state ② is having  $\phi = 1$  that means it is saturated.

$\therefore P_s = P_g$  for state ②

from table  $P_g$  at  $6^\circ\text{C} = 0.009346 \text{ bar}$

$$\therefore \omega_2 = \frac{0.622 \times 0.009346}{(1.01326 - 0.009346)}$$

$$\therefore \omega_2 = 0.00579$$

$$\text{Sr, } \dot{m}_{s_2} = 0.00579 \times \dot{m}_s$$

$$\text{ng Mass of condensation} = \dot{m}_{s_1} - \dot{m}_{s_2}$$

$$= (0.007213 - 0.00579) \times \dot{m}_g$$

$$= 0.001423 \times 0.6017 \times 3600$$

$$= 3.082 \text{ kg/h} //$$